

A Microscope Dark Field Illuminator Using a Pixel Ring

In some scientific disciplines, eg mineralogy and gemmology microscopic observations are best made with illumination coming in from the sides and with a dark field behind the subject. In the past this was done with an incandescent lamp with appropriate baffles to control the illumination field. This works but has the disadvantage of getting very hot, sometimes dangerously hot.

The advent of very bright LED's in the Neopixel format opens up new possibilities. The first working model of this approach is shown in figure 1 below.

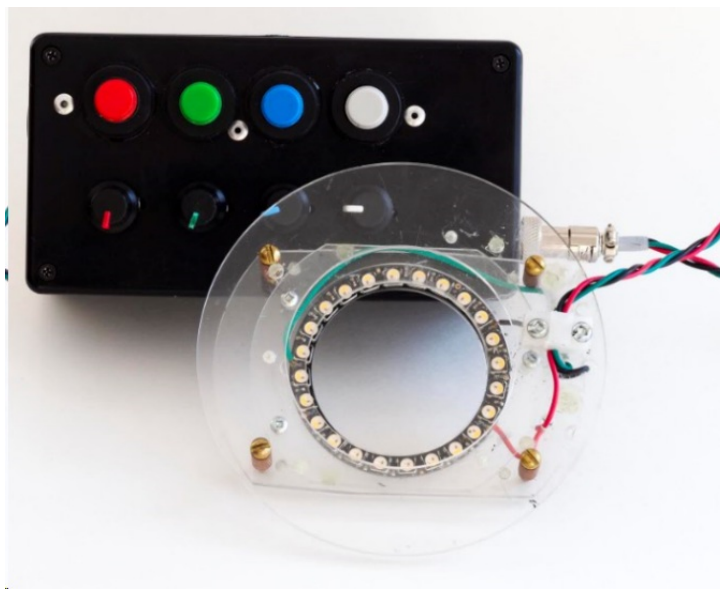


Figure 1

It consists of a neopixel ring mounted in a holder assembly which goes on the microscope stage and a control box containing four on/off pushbuttons and four potentiometers which control the brightness for each of the red, green, blue and white colours plus an Adafruit 5volt Pro trinket controller. The holder assembly shown will be replaced with one made of new, black plastic.

This assembly is shown below in figure 2 mounted on the microscope along with the diffuser ring which controls the illumination field.



Figure 2

The control box is shown open with the Trinket and other components visible in Figure 3 below.

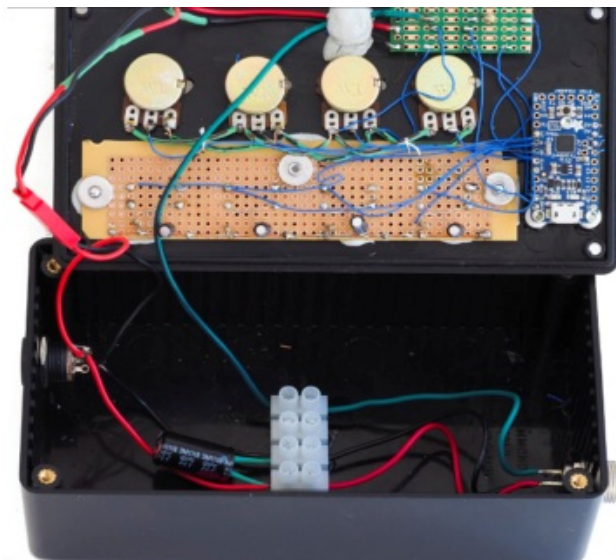


Figure 3

Figure 4 below shows the type of image produced by this type of illumination.

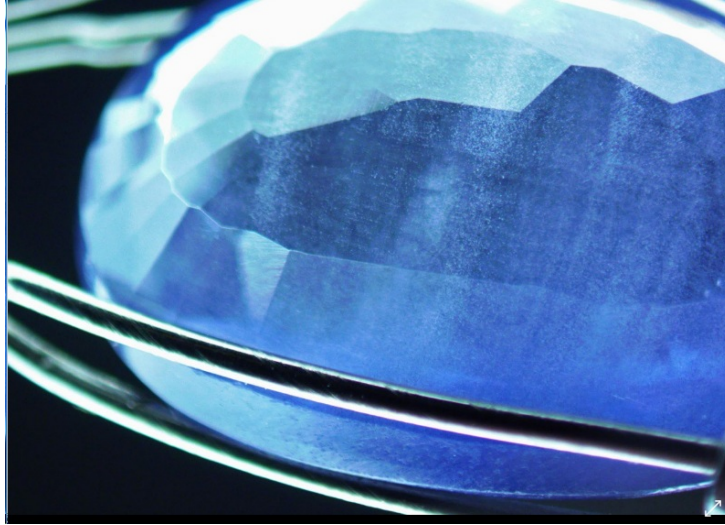


Figure 4

The subject here is a man made cut sapphire with clouds of tiny bubbles shown.

The software for this project is quite simple using the neopixel library and pushbutton and analog pot brightness control. The pb's are push on/push off so obviously need debouncing. This was done by configuring the inputs as "input_pullup" with a 5 mfd capacitor connected input to ground. This gives 100 ms time constant assuming a 20k pullup resistor, plenty to block any bouncing of the pushbuttons.

The availability of good colour illumination is a new feature and may or may not be useful. It is being explored now. There is also the possibility illumination using a limited sector of the ring but exploration of this is in the future.